REVIEW

TUMORS OF THE NERVOUS SYSTEM

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DIFFERENTIAL OF IMAGING CHARACTERISTICS OF PRIMARY BRAIN TUMORS AND METASTASES

- Metastases
  - Primary can not be differentiated from Mets
  - Multiple ring enhancing lesions
  - History of malignancy elsewhere makes this more likely
- Abscess
  - Usually can not differentiate by imaging
  - Clinical presentation of fever, elevated WBC
- Stroke/ hematoma
  - May conform to a vascular distribution

TERMS:

Glioma: Refers to all glial tumors in general but is often also used to mean astrocytoma

Intra-axial: Lesion is in brain or spinal cord tissue, e.g. astrocytoma and oligodendroglioma

Extra-axial: Lesion is extrinsic to brain, e.g. meningioma and schwannoma

Anaplasia: Cellular atypia and loss of differentiation associated with malignant tumors

CNS TUMORS IN A NUTSHELL

- 2.4% (ca 13,000) of all cancer deaths are due to CNS tumors
  Intracranial tumors: 10-17: 100,000; intraspinal tumors: 1-2: 100,000
- The incidence has a peak in childhood and then declines till 25 years; the CNS is among the three leading sites of cancer mortality in the first three decades of life
- In children brain tumors are the most frequent neoplasm after leukemia, accounting for about 20% of all cancers , but the most common CNS cancer killers.
- In adulthood there is a continuous increase with advancing age
- An apparent rise in the incidence in recent years has be attributed to more accurate diagnosis, especially with MRI imaging; some investigators argue that the rise is real
- Inarguably the incidence of cerebral lymphoma has increased
- Most common primary brain tumors are meningioma, astrocytoma, and glioblastoma multiforme (aka. anaplastic astrocytoma WHOgrade IV)

CNS TUMORS IN A NUTSHELL

- In children primary brain tumors are more common than metastatic tumors
- In adults metastatic brain tumors are more common than primary brain tumors
- The most common brain tumors in children are pilocytic astrocytoma and medulloblastoma
- Most brain tumors in children are infratentorial, located in the cerebellum and brainstem
- Most brain tumors in adults are supratentorial, located in the cerebrum
- the most common tumors of the spinal cord are schwannoma, meningioma, and ependymoma

CNS TUMORS IN A NUTSHELL

UNIQUE CHARACTERISTICS

Histology:

Morphologic differences between benign and malignant tumors are more subtle than in other organs.

Biologic behavior:

Well-differentiated low-grade lesions with low mitotic (= growth) rate and uniform cellularity may destroy important areas of brain and lead to significant morbidity and mortality and poor prognosis

Anatomic site:

Regardless of the type and differentiation of the tumor, in certain locations (e.g. medulla oblongata) it may cause death by compressing vital centers

Pattern of spread:

Extracranial spread of intracranial tumors is extremely rare
Metastatic spread to the CNS from extracranial tumors is common

GENETIC TUMOR SYNDROMES

• Account for a small proportion (1%- 4%) of CNS tumors

VON RECKLINGHOUSEN NEUROFIBROMATOSIS VRNF- Neurofibromatosis type 1-NF1

• Autosomal dominant inheritance pattern
• Caused by mutations of a gene on chromosome 17q that encodes neurofibromin
• Neurofibromin is involved in control of cell proliferation and acts as a tumor suppressor of Ras oncoprotein

BILATERAL ACOUSTIC NEUROFIBROMATOSIS - BANF - Neurofibromatosis type 2-NF2

• Autosomal dominant inheritance pattern
• Acoustic and spinal Schwannomas

TUBEROUS SCLEROSIS

• Autosomal dominant inheritance pattern
• Clinical characteristics:
    mental retardation, seizures
    subependymal giant cell astrocytomas, cortical hamartomas (tubers) of the CNS
    tumors and hamartomas of other organ
•Two mutations:
    TS1 on 9q which encodes hamartin
    TS2 on 16p which encodes tuberin - acts as a tumor suppressor

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GENETIC TUMOR SYNDROMES CONTINUED

VON HIPPEL LINDAU DISEASE

- Autosomal dominant inheritance pattern
- Associated with:
    - hemangioblastomas of the cerebellum and retina
    - cysts of the liver and pancreas
    - pheochromocytomas
    - kidney tumors
    - pancreatic tumors
- Linked to *VHL*, a tumor suppressor gene on chromosome 3p
- pVHL is involved in the rate of mRNA transcription*
- pVHL inhibits vascular endothelial growth factor (VEGF)

*During transcription, cellular proteins can intervene in one of three stages -- initiation, elongation, and termination. VHL blocks the protein that stimulates elongation.

IMPORTANT FACTORS
IN THE PATHOGENESIS OF BRAIN TUMORS

AGE:

- **Cerebellum is the most frequent site of CNS tumors children, cerebrum is the most frequent site of CNS tumors in adults.**

- **Embryonal tumors of the brain (cerebellar medulloblastoma) and other organs (adrenal neuroblastoma) occur predominantly in children.**

- **Explanation:**
  - **Neurogenesis and neuronal migration in the cerebrum are largely completed by midgestation, but in the cerebellum they continue for the first year of life.**
  - **Cerebellum is the most cellular part of the CNS (there are more granular neurons than all other neurons in the CNS together) and takes the longest to develop.**
  - **Glial cells are actively produced in cerebellum in childhood and the brisk cell division allows for new gene defects to arise and for inherited defects to be unmasked.**

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS CONTINUED

SEX:

- **Males have more CNS tumors overall**
- **Females have more meningiomas**

RADIATION:

- **An increased incidence of CNS tumors, especially meningiomas, has been reported in patients who have received even low-dose radiation to the head.**
- **Children with ALL who have been treated with craniospinal radiation are at high risk for developing meningiomas and gliomas.**
- **The frequency of PNETs, including medulloblastoma, is increased in children, whose mothers had abdominal x-rays during pregnancy.**
- **Tumors emerge many years after radiation has been given and their incidence appears to be rising.**

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS
CONTINUED

CHEMICAL CARCINOGENS:

- Nitroso compounds (NOCs)
  - The most potent neurocarcinogens in experimental animals
  - Present in foods (cured meats, fish, vegetables), cosmetics, rubber products, beer and water
  - Synthesized in the mouth, stomach, and bladder by nitrosation of amines and amides in the diet, therefore ubiquitous
- Formaldehyde
  - Pathologist and funeral directors have more CNS cancers than the general population
- Vinyl chloride
  - Plastics manufacture
- Acrylonitrile
  - Textiles and plastics manufacture

IMPORTANT FACTORS IN THE PATHOGENESIS OF BRAIN TUMORS
CONTINUED

IMMUNOSUPPRESSION:

Cerebral lymphomas, usually B-cell, are frequent in patients with :

- AIDS
- Renal transplants
- Congenital immunodeficiency syndromes
- Immunosupressed states

EBV-INFECTION:

Some CNS lymphomas contain EBV-DNA

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DIAGNOSIS OF CNS TUMORS

CLINICAL PRESENTATION:
- General symptoms and signs
- Thorough neurologic examination

IMAGING STUDIES:
- MRI
- CT

LABORATORY EVALUATION:
- Requires brain biopsy
- Pathological diagnosis is based on gross, microscopic (growth pattern, vascularity, special features, such as rosettes), immunohistochemical, and ultrastructural features
- Assessment of histologic grade is based on:
  - Cellularity
  - Nuclear atypia
  - Proliferative index (mitotic count or Ki-67, which identifies proliferating cells, even if they are not in mitosis)
  - Necrosis
- Chromosomal and molecular analysis

MOLECULAR MARKERS IN CNS TUMORS

IDH **(*IDH 1* and *IDH2* isocitrate dehydrogenase) gene mutations**
- **increased production of 2-hydroxyglutarate**
- **Present in low grade astrocytomas and oligodendrogliomas**

Mutations in the telomerase promoter gene *hTERT*:
- **In astrocytomas including glioblastomas**
- **Uncommon tumors with *IDH* mutations, which have LOF in *ATRX* (normally suppresses alternative lengthening of telomeres)**

Co-deletion of 1p and 19q chromosomal segments:
- **Oligodendrogliomas**

Loss of 22q:
- **meningioma**
- **Schwannoma (also mutations in the *NF2* gene)**
- **atypical teratoid-rhabdoid tumor**

PTCH1 **mutations :**
- **Important in cerebellar granular cell tumors (e.g. medulloblastoma)**

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PTCH1 is also implicated in nevoid basal cell carcinoma syndrome and other skin tumors, esophageal squamous cell carcinoma and transitional cell carcinomas of the urinary tract.

MOLECULAR MARKERS IN CNS TUMORS CONTINUED

Other genetic alterations commonly found in tumors outside the CNS; the more malignant the tumor (e.g. GBM), the more mutations

- **Loss of heterozygosity (LOH) 10q (60-90%); in GBMs but not in lower grades**
- ***EGFR* amplification (40-50%); more common in primary GBM**
- ***MDM2* amplification/overexpression (10-15% of GBM)**
- ***P53* mutations or deletion (20-40%)**
- ***RB* mutations or deletion**
- ***P16INK4a* mutation**
- ***PTEN* mutation**
- ***N-Myc* mutation**
- **Up-regulation of *cyclin D2***

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CLASSIFICATION OF BRAIN TUMORS

TUMORS OF NEUROGLIAL CELLS

A. TUMORS OF GLIAL CELLS
- Astrocytoma
- Glioblastoma multiforme (anaplastic astrocytoma, grade IV)
- Oligodendroglioma
- Ependymoma - choroid plexus papilloma

B. NEURONAL TUMORS
- Ganglioglioma
- Gangliocytoma
- Central neurocytoma

C. EMBRYONAL TUMORS
- Medulloblastoma and other CNS PNETs

TUMORS OF CRANIAL AND SPINAL NERVES
- Schwannoma
- Neurofibroma

MENINGEAL TUMORS
- Meningioma

MESENCHYMAL TUMORS
- Sarcoma
- Hemangioblastoma

CEREBRAL LYMPHOMAS

GERM CELL TUMORS
- Teratoma
- Germinoma
- Craniopharyngioma

TUMORS OF THE PITUITARY GLAND
- Pituitary adenoma

METASTATIC TUMORS

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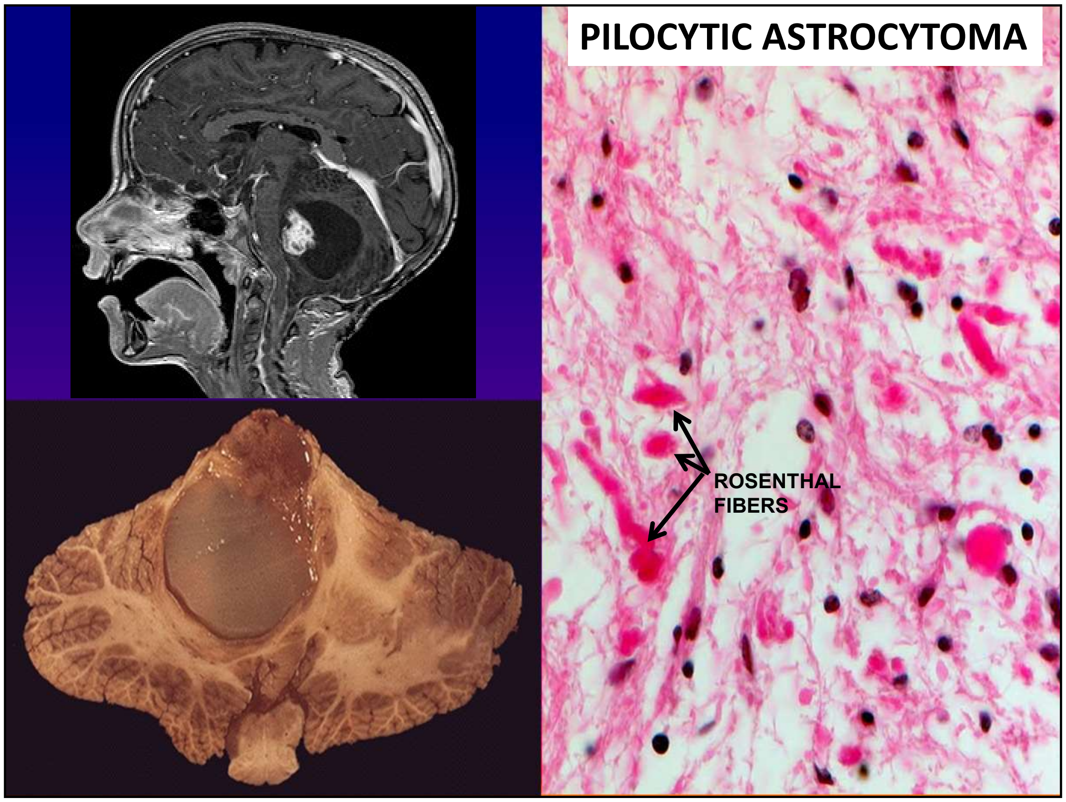
WHO GRADING SYSTEM FOR ASTROCYTOMAS

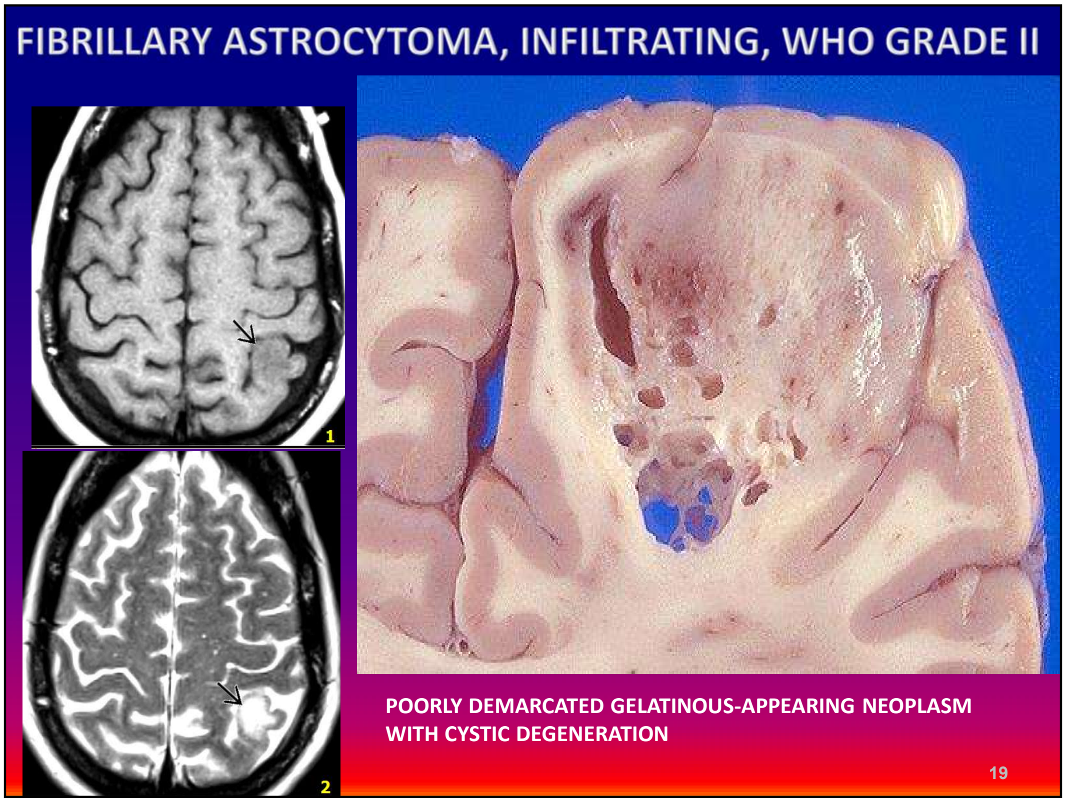
Grade I - Pilocytic astrocytoma:
Benign cytological features

Grade II - Low-grade astrocytoma:
Moderate cellularity, no anaplasia or mitotic activity

Grade III - Anaplastic astrocytoma:
Increased cellularity, nuclear atypia, mitoses

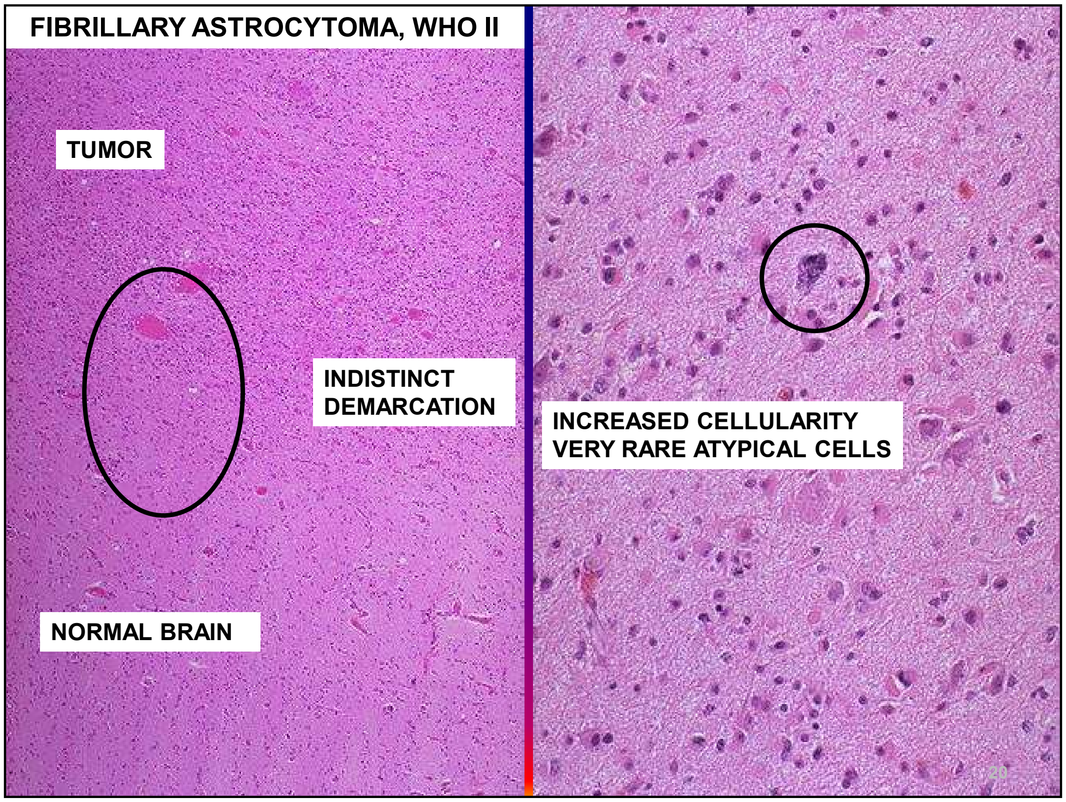
Grade IV – Glioblastoma multiforme:
Same as Grade III plus vascular endothelial proliferation and necrosis with pseudopalisading pyknotic tumor cell nuclei surrounding it


PILOCYTIC ASTROCYTOMA
ROSENTHAL
FIBERS

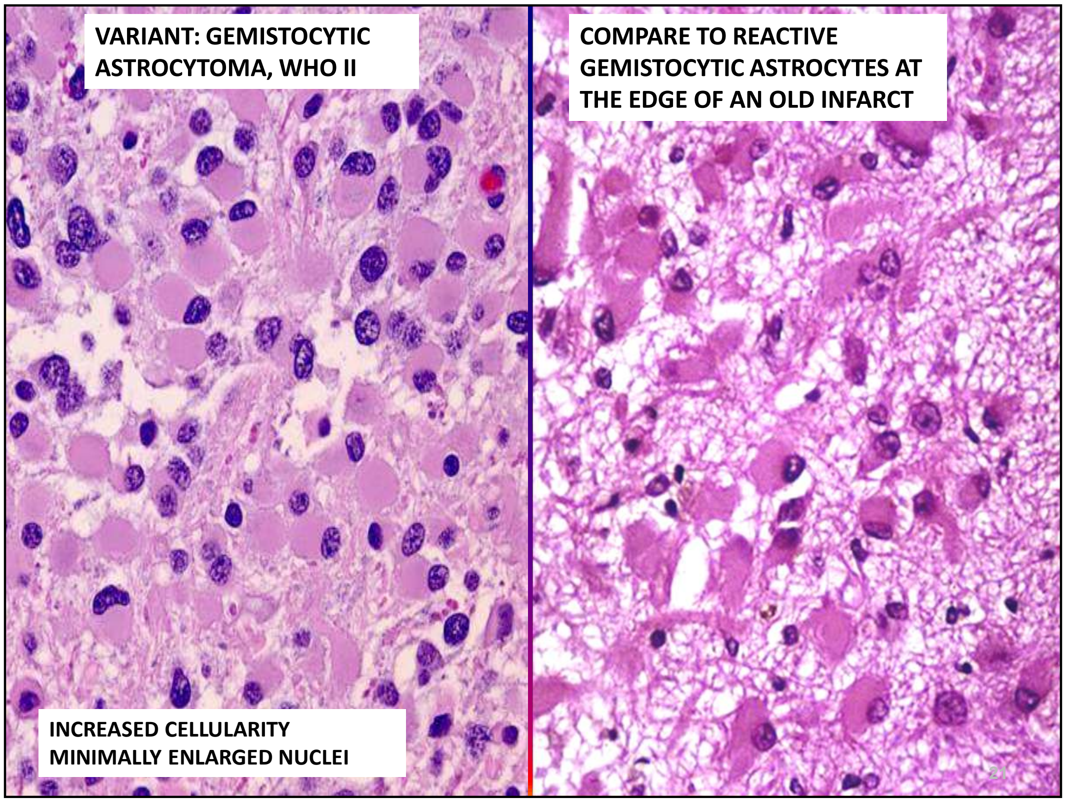


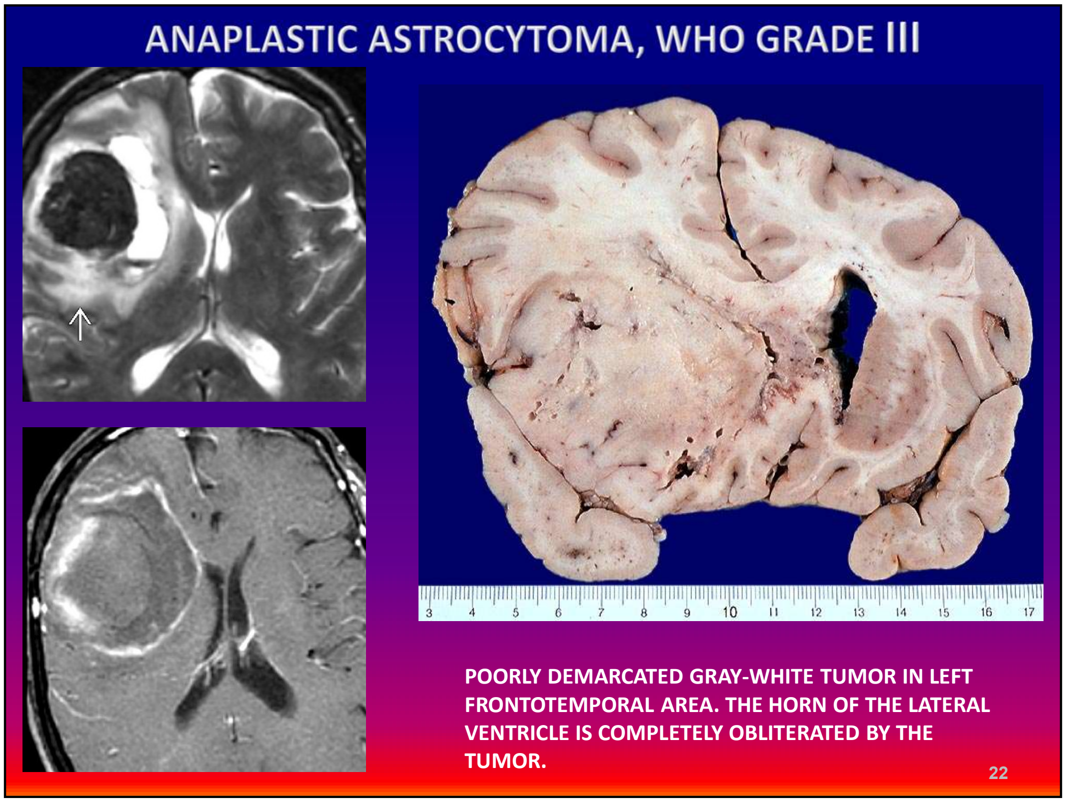
Left: Astrocytoma who grade 2. Left parietal lobe MRI T1 and T2 images.
Right: Fibrillary Astrocytoma, poorly demarcated gelatinous mass with cystic degeneration

- Astrocytomas can occur anywhere but are most frequent in the cerebral hemispheres.
- Most are poorly demarcated with borders difficult to determine during surgery.
- Difficult to differentiate histologically from reactive gliosis in frozen sections.

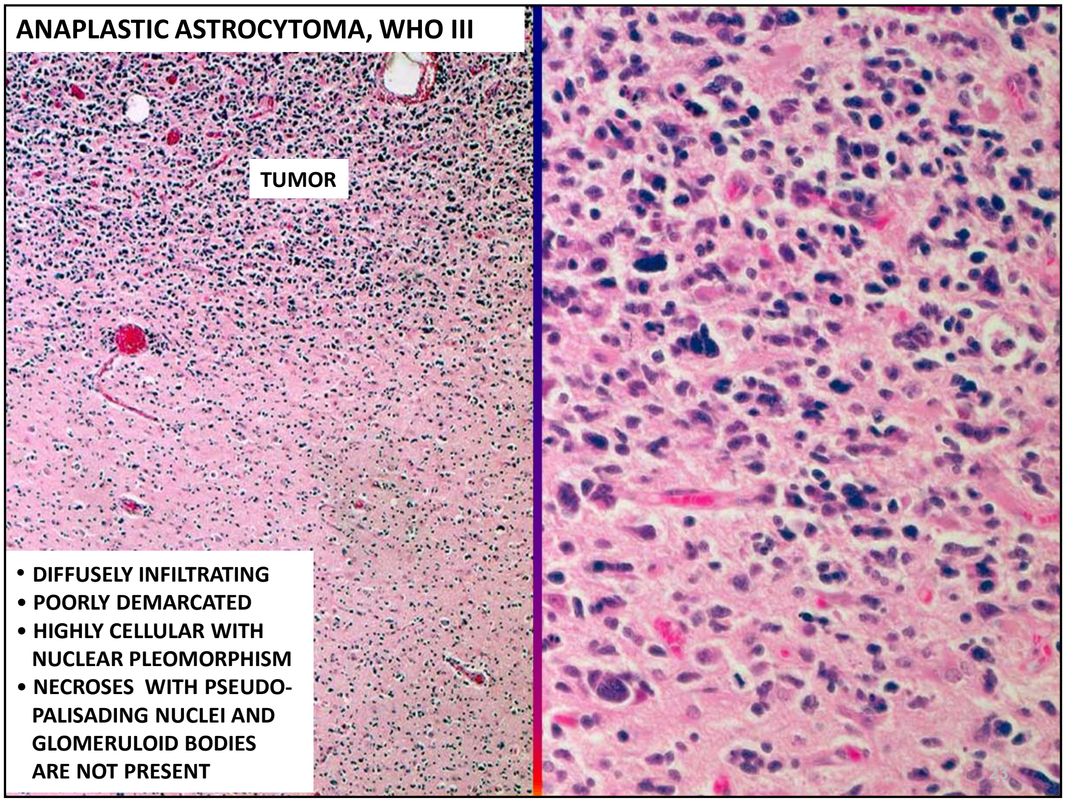


Low grade infiltrating astrocytoma (glioma) on top, normal brain on bottom. Note increased cellularity and indistinct margin of the tumor.


VARIANT: GEMISTOCYTIC ASTROCYTOMA, WHO II
INCREASED CELLULARITY
MINIMALLY ENLARGED NUCLEI
COMPARE TO REACTIVE GEMISTOCYTIC ASTROCYTES AT THE EDGE OF AN OLD INFARCT



Left: Grade 3 Astrocytoma. T2 and T1 post contrast enhancement images. Hemorrhage in the T2 image is causing signal void (black area arrow)



Clinical findings: 44 y o woman with a recurring brain tumor. **Pathologic diagnosis:** Anaplastic astrocytoma, WHO grade III. **Description:** Diffusely infiltrating poorly demarcated highly cellular tumor composed of pleomorphic cells with atypical nuclei. Stain: HE.

GLIOBLASTOMA MULTIFORME (GBM)
ANAPLASTIC ASTROCYTOMA, WHO IV

- The most malignant glioma.
- Most common in middle aged or older adults, but may occur at any age.
- Most common sites are the frontal and temporal lobes, but GBM may involve any part of the CNS.
- Most GBMs arise de novo (primary GBM), some arise by malignant transformation from low-grade astrocytomas (secondary GBM).
- Pseudopalisading of tumor cells around central necrosis and vascular proliferation are diagnostic features.
- Primary GBMs are often composed of small undifferentiated cells (small cell glioblastoma), show extensive ischemic necrosis and have a higher proliferative index.
- Primary GBMs are more common in older patients and are more aggressive.
- Secondary GBMs are composed of larger cells with astrocytic differentiation.
- Survival exceeding one year is rare in either primary or secondary GBM.
- Postoperative irradiation and chemotherapy prolong survival minimally.

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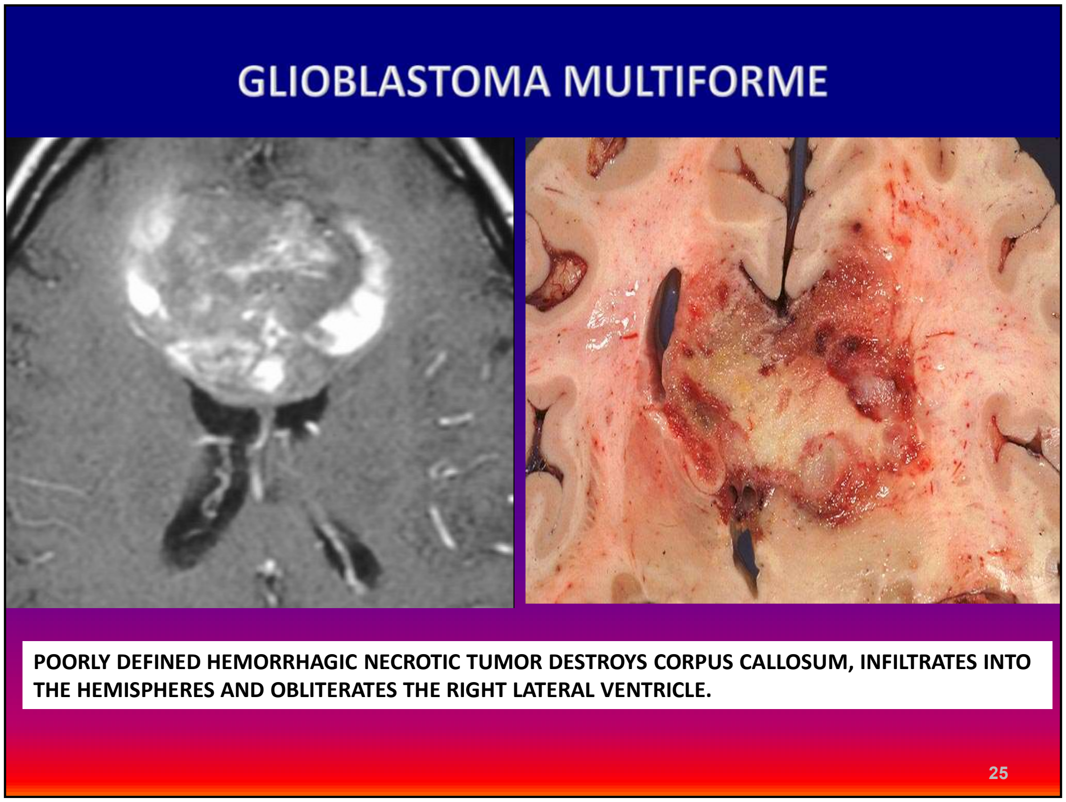

52 yo woman. Clinical findings:. Neurologic symptoms for six months with aphasia, psychomotor slowing, and urinary incontinence. Imaging studies revealed a mass destroying corpus callosum. One month later the patient's status grew significantly worse with general symptoms of fever and malaise. Immediate cause of death was aspiration pneumonia.

ANGIOGENESIS IN GBM

- **Hypoxia develops as GBM outgrows its vascular supply.**
- **Hypoxia induces upregulation of hypoxia inducible factor 1 (HIF-1).**
- **HIF-1 in turn stimulates the expression of vascular endothelial growth factor (VEGF).VEGF stimulates formation of new vessels, which allows continuing tumor growth.**
- **The new vessels are often arranged in glomeruloid formations and lack a blood-brain barrier, which contributes to cerebral edema, a clinically important feature of GBM.**
- **Anti-angiogenic treatment can temporarily suppress the tumor**

MOLECULAR EVENTS IN THE DEVELOPMENT OF GBM

- **Mutation of the tumor suppressor gene *p53* on 17p is a pivotal event in the transformation of normal to neoplastic astrocytes.**
- **Secondary GBMs are characterized by early *p53* mutation and subsequent loss of the long arm of chromosome 10.**
- **Deletion of the tumor suppressor *PTEN* (Phosphatase and Tensin Homologue) in chromosome 10 with losses of several other tumor suppressor genes in other chromosomal loci convert low-grade astrocytomas to anaplastic astrocytomas and GBMs.**
- **Overexpression of the epidermal growth factor receptor (EGFR).**
- **Primary GBMs overexpress epidermal growth factor receptor (EGFR), a tyrosine kinase coded by a gene on 7p.**
- **Expression of *EGFR* and related genes determines the response of GBM to tyrosine kinase inhibitors, which are used in GBM chemotherapy.**

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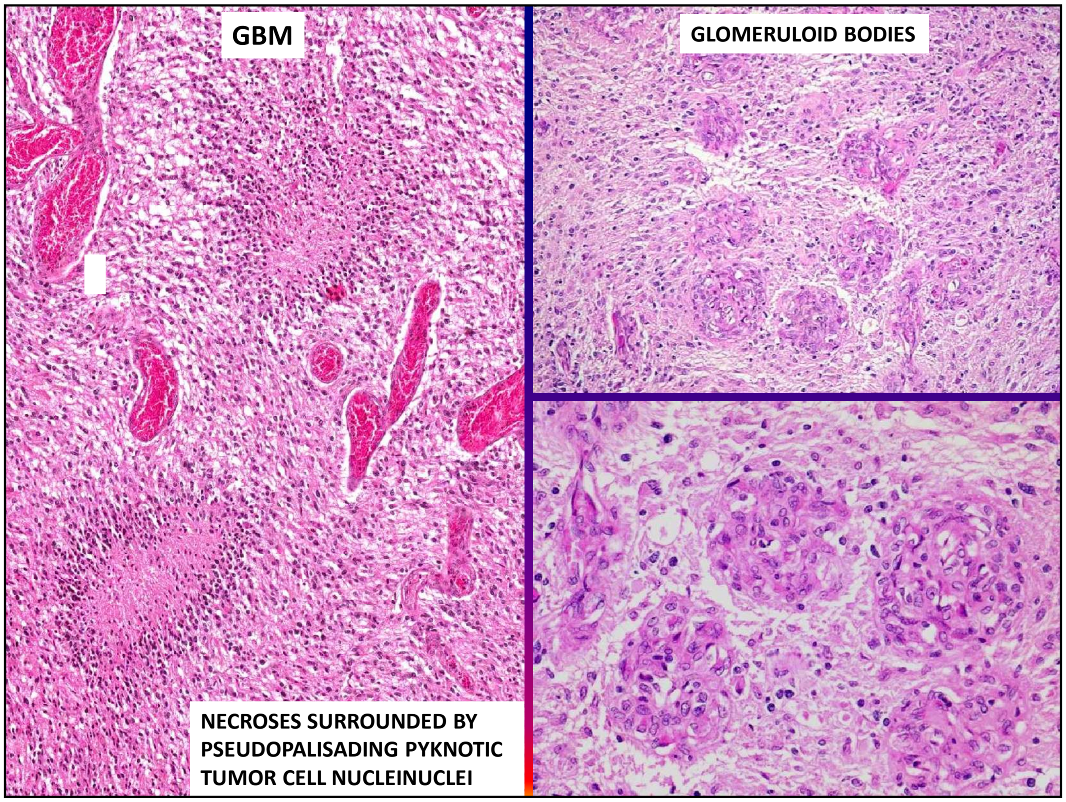


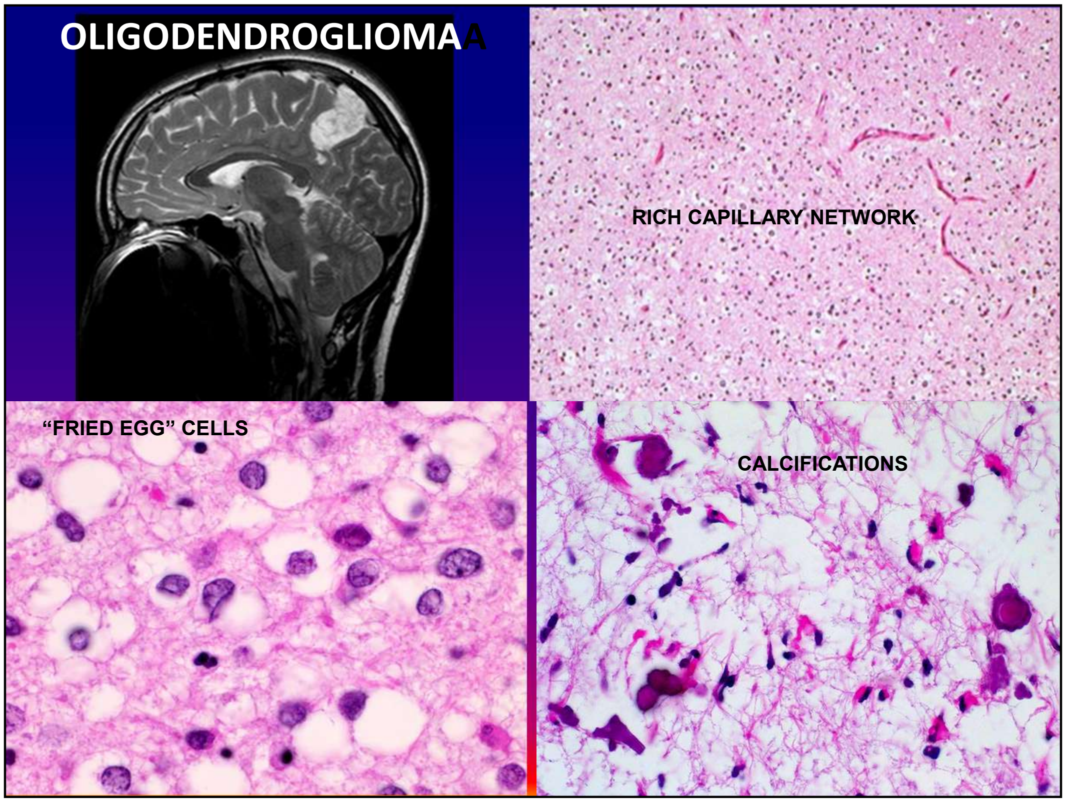
GLIOBLASTOMA MULTIFORME:

- Highly cellular malignant tumor
- H

ecroses surrounded by pseudopalisading tumor cells and glomeruloid bodies are a useful diagnostic feature

ASTROCYTOMAS RECAP

- Account for 80% of adult primary brain tumors
- Most frequent in the 4thTH- 6TH decades
- Usually located in the cerebral hemispheres
- Clinical symptoms include headache and seizures; focal symptoms are determined by the location
- Histologic differentiation correlates with clinical course and outcome
- Well-differentiated astrocytomas grow slowly with a mean survival >5 years
- All astrocytomas have malignant potential and often develop more malignant features later during the course.
- Many astrocytomas present from the start as anaplastic astrocytoma, grade IV, aka. glioblastoma multiforme (GBM).
- GBMs have a mean survival rate of 8-10 months with <10% 2-year survival rate.



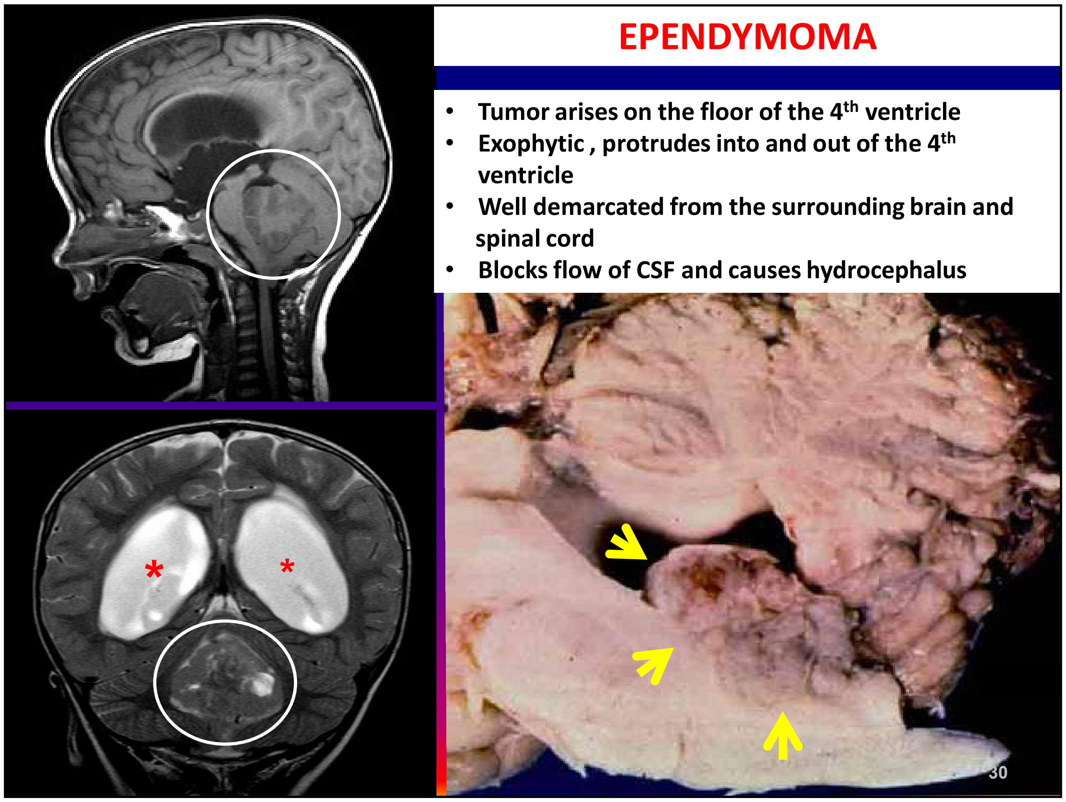
OLIGODENDROGLIOMA

- 5-15% of gliomas.
- Usually located in the cerebral hemispheres.
- Most common in the 4th and 5th decade.
- Insidious, slow growing.
- Mean survival 5 years.
- Appears more circumscribed than astrocytomas, but infiltrates the cortex diffusely.

- Often develops areas of cystic degeneration and bleeds because of rich capillary network.
- Calcifications are common in the periphery of the tumor and help in radiologic diagnosis; seen on CT but may not be seen on MRI.

chemosensitive

Molecular events:

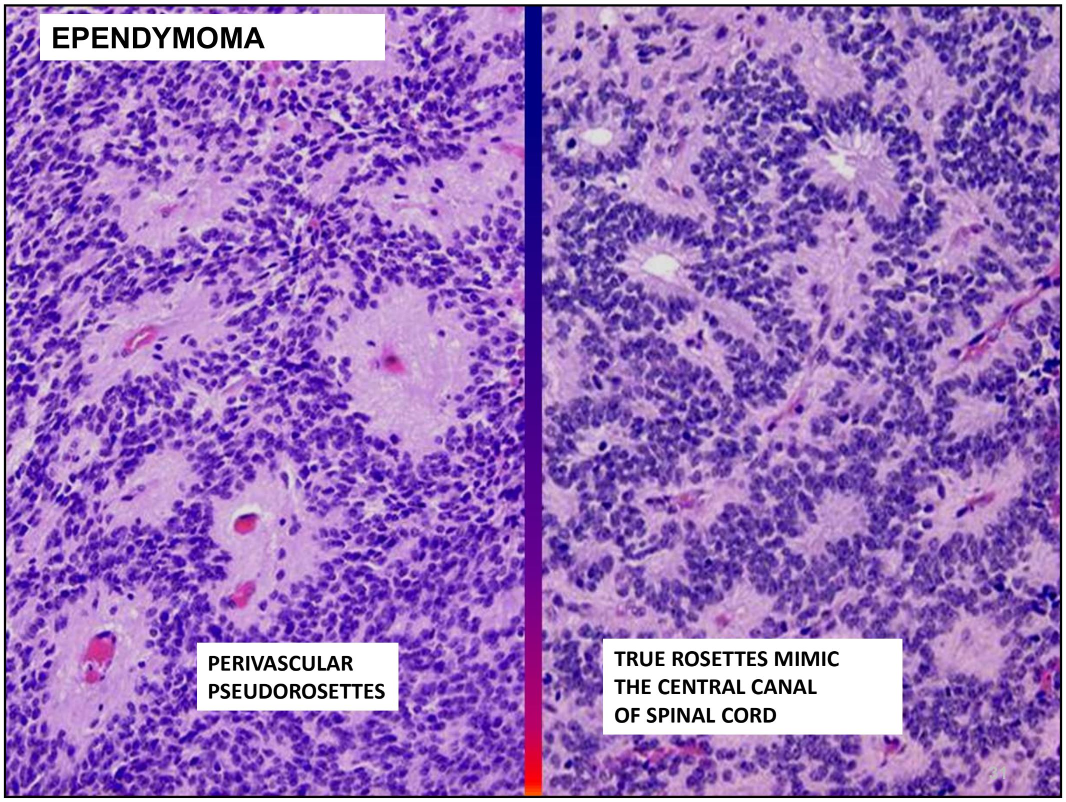


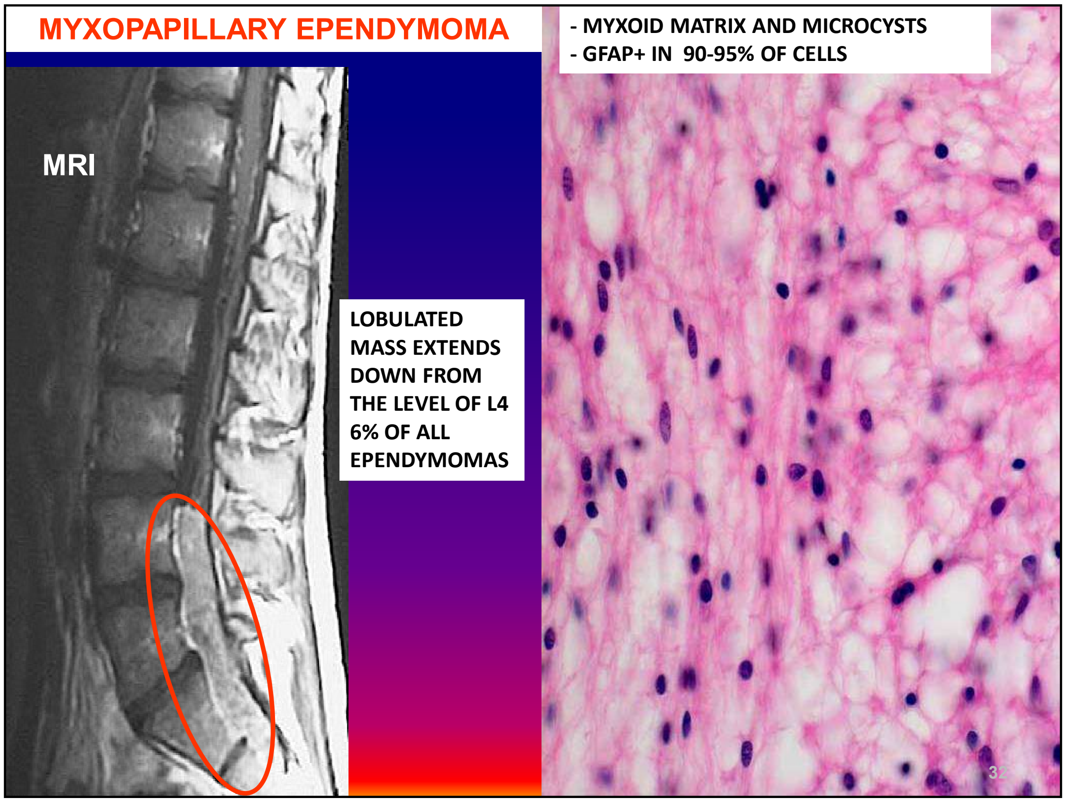
EPENDYMOMA

Tumor of children and young adults.

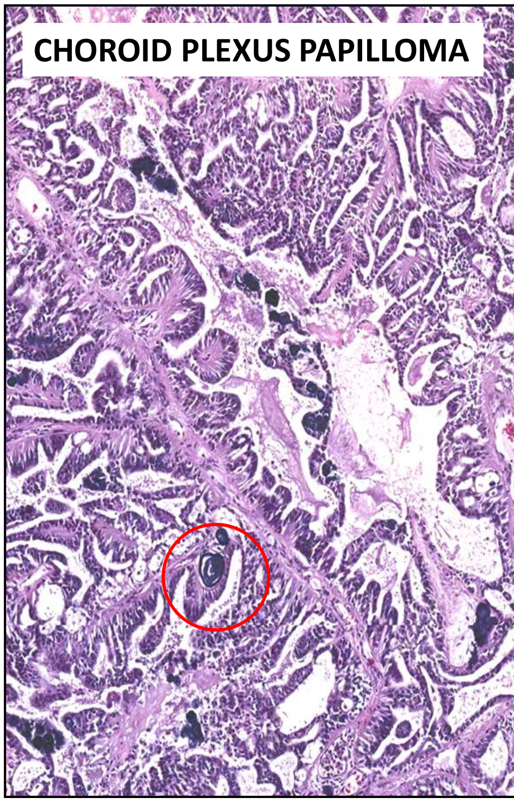
fourth ventricle

hydrocephalus


EPENDYMOMA
PERIVASCULAR
PSEUDOROSETTES
TRUE ROSETTES MIMIC
THE CENTRAL CANAL
OF SPINAL CORD



Diagnosis: Myxopapillary ependymoma. **Description:** Myxoid matrix and microcysts are seen between tumor cells. **Additional findings:** Ca. 90-95% of tumor cells stain for GFAP. The proliferation rate of this tumor is extremely low, far below 1%. **Clinical findings:** Spinal cord ependymoma in filum terminale. **Comment:** Myxopapilary ependymomas make up 6%of all ependymomas and they are practically always found in the conus- cauda equina-filum terminale region. **Stain:** HE.



CHOROID PLEXUS PAPILLOMA

- Most common in children
- Occurs anywhere along the choroid plexus
- In the lateral ventricles in children, in the fourth ventricle in adults
- Often has psammoma bodies (lamellar calcifications)
- Papillary growths recapitulate the structure of the normal choroid plexus almost exactly: fibrovascular stalks are covered with a cuboidal or columnar epithelium
- Clinically present with hydrocephalus due to obstruction of the ventricular system by tumor or overproduction of CSF

CHROID PLEXUS CARCINOMA

- Resembles adenocarcinoma
- Rare, more common in children than adults
- Important to differentiate from metastatic carcinoma in adults

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PRIMITIVE NEUROECTODERMAL TUMORS (PNETs)

•The term primitive neuroectodermal tumor (PNET) has been applied to medulloblastoma and other "small blue round cell tumors" of the brain

MEDULLOBLASTOMA,

- The second most frequent CNS tumor in children after pilocytic astrocytoma
- Occurs in the first decade of life with a second peak in the early 20s
- Embryonal tumor, analogous to Wilms tumor and adrenal neuroblastoma (“small blue round cell tumor”)
- High incidence in infants and children
- Undifferentiated, immature appearance; resembles developing neural tissue
- Highly malignant
- Cells of origin: primitive cells of the external granular layer of the cerebellum, which persists until the beginning of the second year of life
- External granular layer of the cerebellum:
  - Made up of precursor cells that migrate from the roof of the fourth ventricle to the surface of the developing cerebellum where they divide and differentiate
  - Neurons move inwards and form the permanent granular layer of the cerebellar cortex.
  - Remnants of the provisional granular layer are seen in the posterior and anterior medullary velum.

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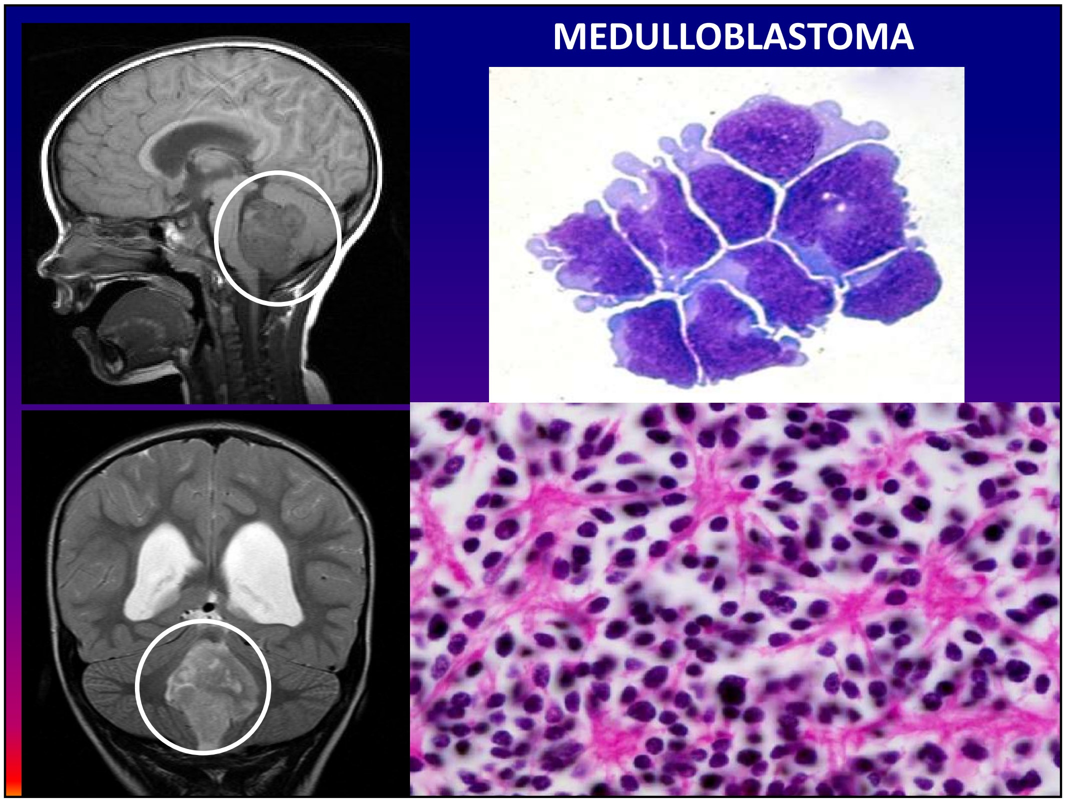


MEDULLOBLASTOMA

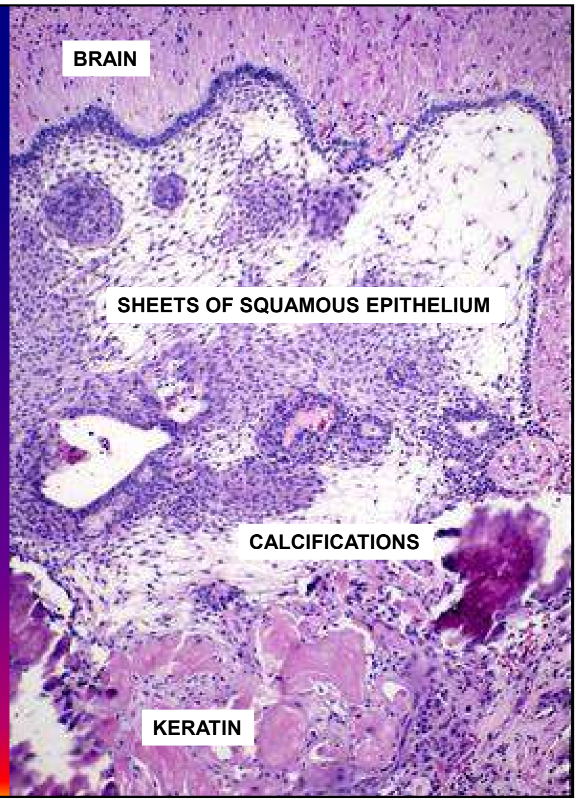
- **Neuronal differentiation is manifested by neuropil and rosette formation**.
- **Rosettes** are groups of tumor cells arranged in a circle around a fibrillary center**.**
- Mature neurons may be found in medulloblastomas infrequently.
- **Glial differentiation is shown by GFAP-positive cells.**

Extracranial metastases are rare, but can occur after operation or shunting

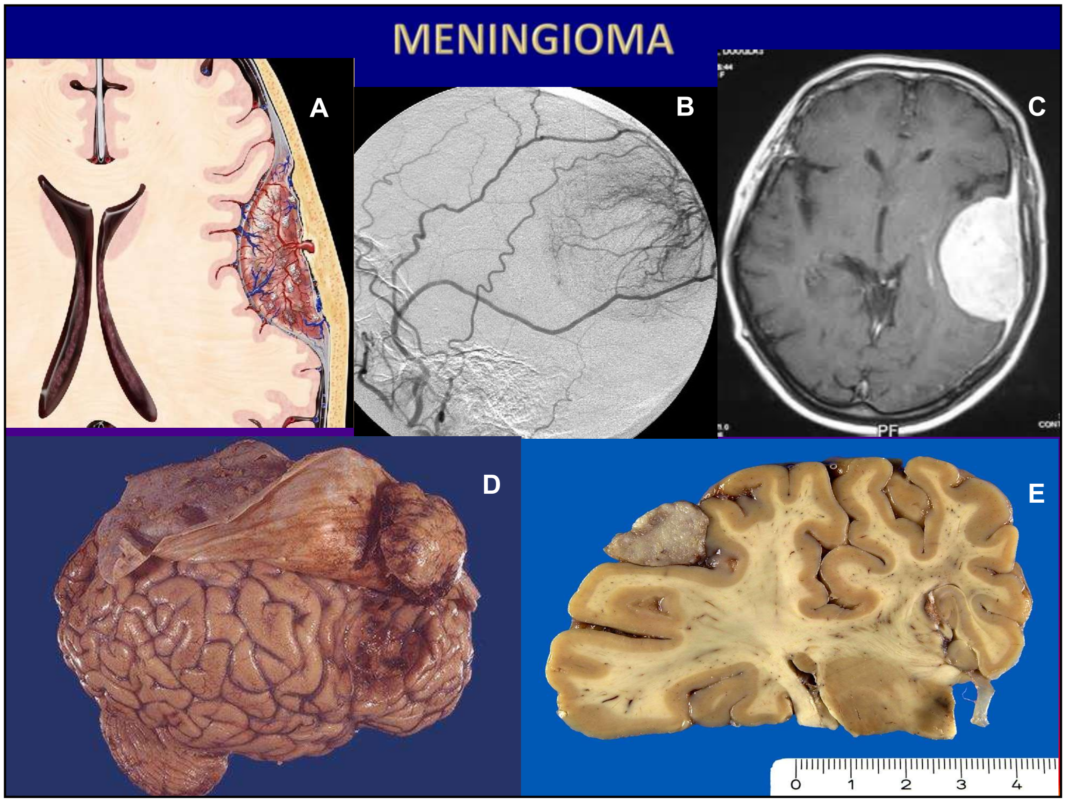
• Lumbar puncture in this setting is contraindicated: can induce cerebellar tonsillar herniation

Focal differentiation into neuronal, oligodendroglial, or ependymal cells may be present.

CRANIOPHARYNGIOMA



BENIGN, SLOW-GROWING TUMOR IN TYPICAL SUPRASELLAR LOCATION
ARISES FROM REMNANTS RATHKE"A POUCH
CYSTS AND CALCIFICATIONS COMMON
DAMAGES ANTERIOR PITUITARY



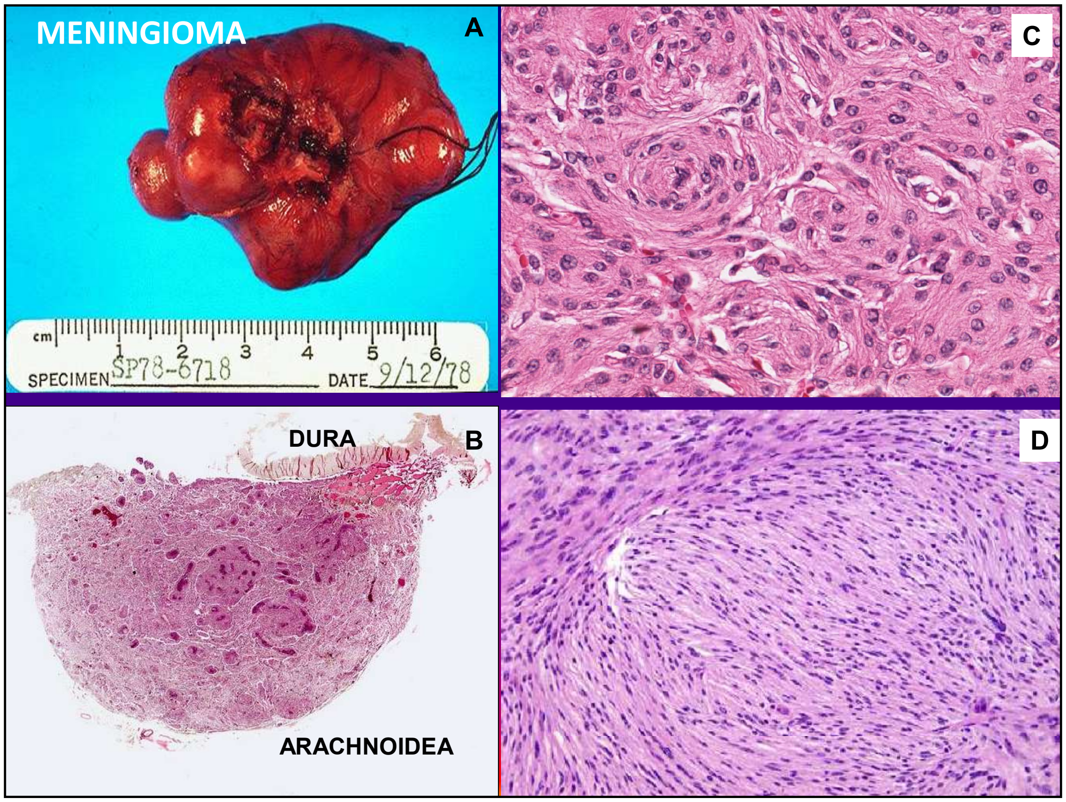
A. Meningioma is an extraaxial mass.
B. Angiogram of external carotid artery demonstrates enhancement
C. MRI with contrast shows dural “tails”
D. Gross brain shows meningioma stuck to the underside of the reflected dura with an impression on the brain
E. Meningioma on the convexity of brain

MENINGIOMA:

estrogen and progesterone

receptors on tumor cells

usually does not invade the brain but displaces it



Meningioma important features: More common in women than men. Has ER/PR receptors and grows faster during pregnancy.

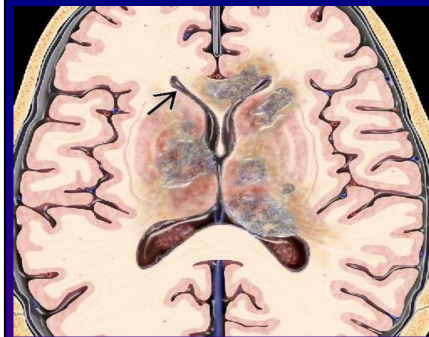
Sharply demarcated tumor attached to dura with classic features of meningioma. The tumor is composed of uniform round cells that form lobules and whorls. Between tumor cells are unremarkable blood vessels and narrow bands of connective tissue.

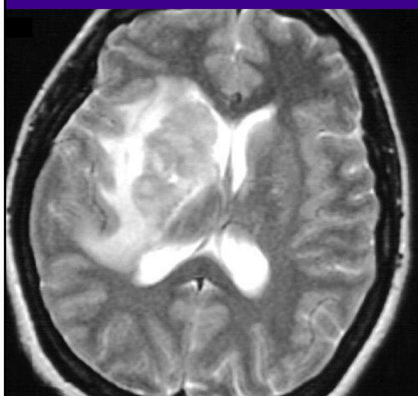
A. Resected meningioma
B. Resected meningioma, subgross
C. Meningothelial meningima, WHO I:

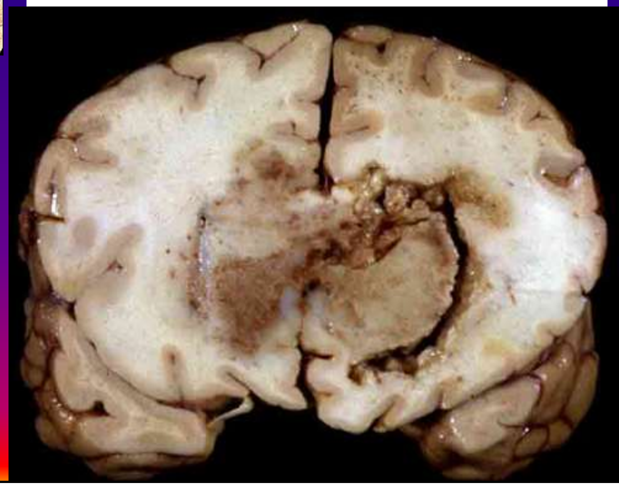
D. Fibroblastic meningioma

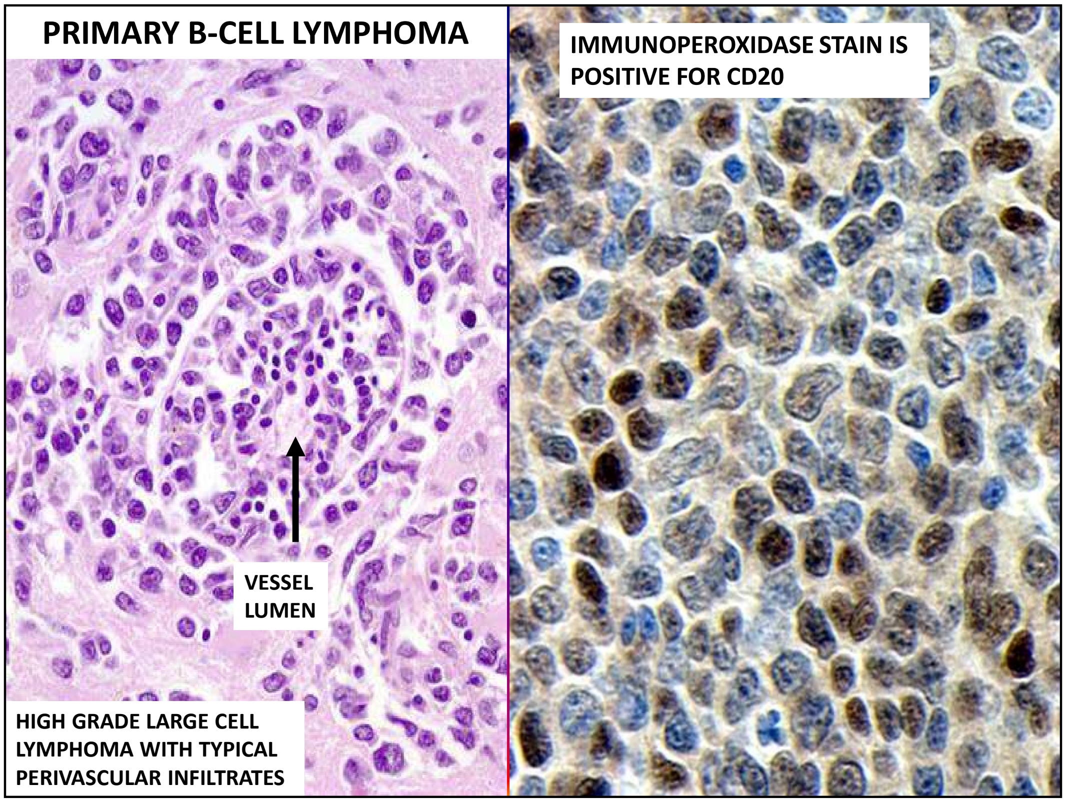
CNS LYMPHOMA

- **Most patients with primary CNS lymphoma have AIDS or are immunosuppressed**
- **Origin: resident CNS lymphoid cells**
- **Grossly destructive like GBM**
- **Besides primary CNS lymphomas, metastatic involvement of CNS by lymphoma is common**



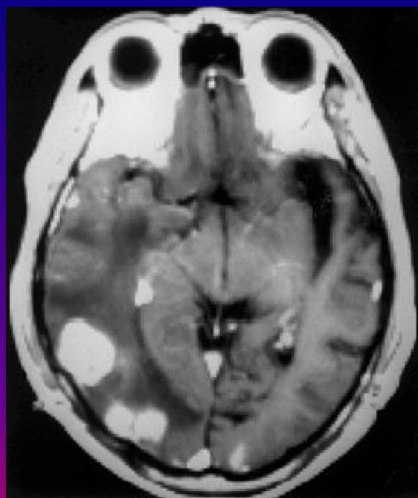




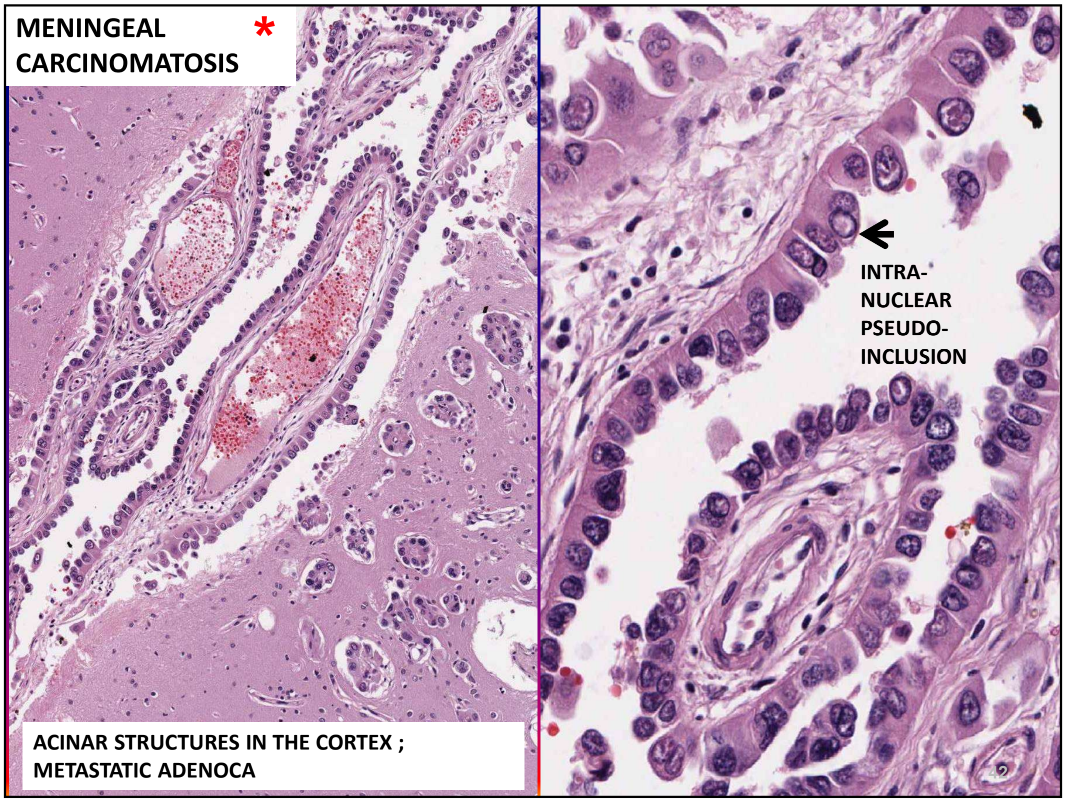


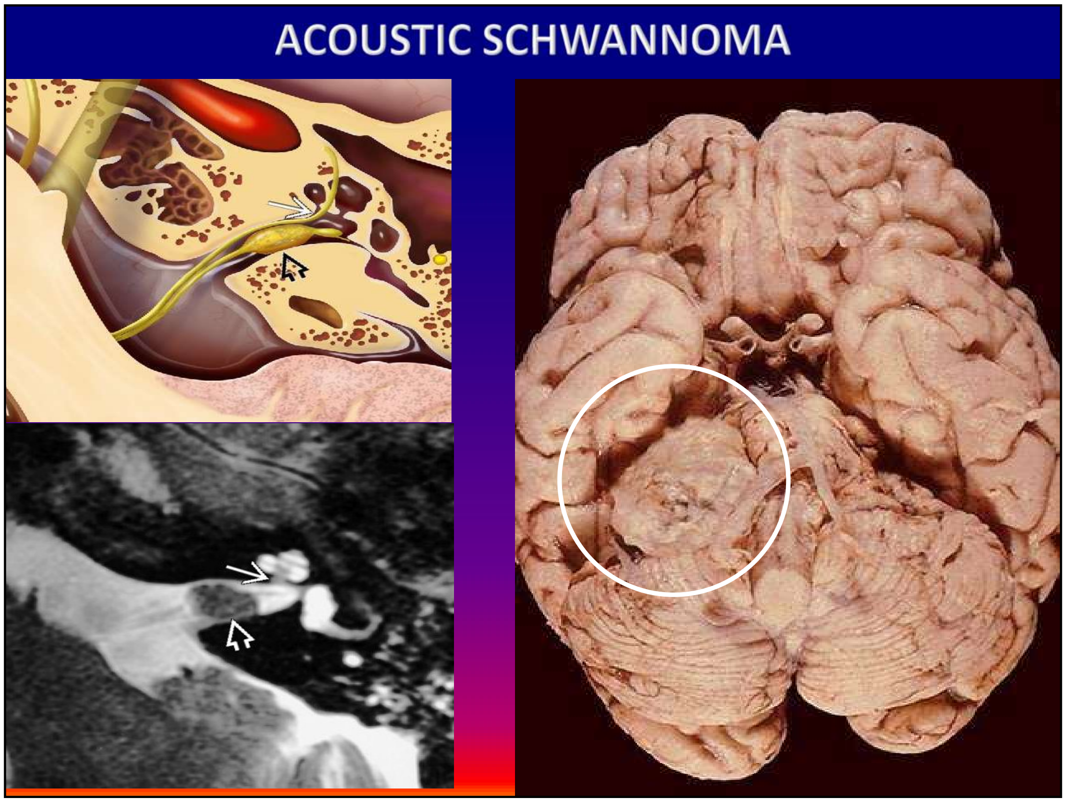
Immunohistochemical studies revealed the tumor cells to be negative for epithelial membrane antigen (EMA), melanocyte marker (HMB-45) and glial fibrillary acid protein (GFAP) and **positive for leucocyte common antigen (LCA/CD45).** The majority of the cells were also **positive for B-cell antigen (CD20).**

METASTATIC TUMORS TO CNS



- Metastasis from another primary site is the most common tumor in the brain in adults
- Brain metastases are found at autopsy in 14-37% of patients with malignant tumors
- Often multiple
- Any tumor can metastasize to CNS
- CNS metastases are most often from lung, (particularly small cell lung cancer) and breast carcinomas, BUT melanoma has the highest frequency
- Up to 1/3 are from unknown primary
- Work-up should include CT of Chest Abdomen and Pelvis
- Lymphoid malignancies (particularly acute lymphoblastic leukemia) metastasize often to the CNS
- Meningeal carcinomatosis is seen in 4-8% of metastatic tumors to the CNS
- CSF: high protein, low glucose, a few lymphocytes and tumor cells

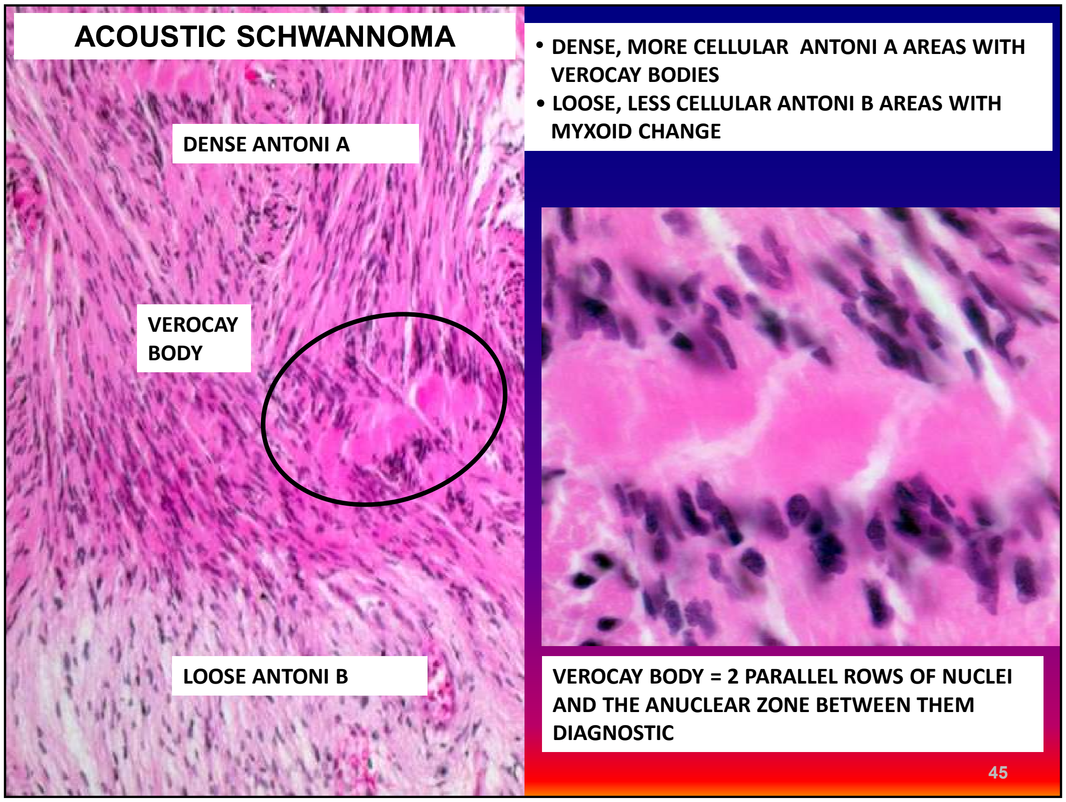

MENINGEAL CARCINOMATOSIS *
ACINAR STRUCTURES IN THE CORTEX ; METASTATIC ADENOCA
INTRA-NUCLEAR PSEUDO-INCLUSION



Acoustic Schwannoma 5mm mass

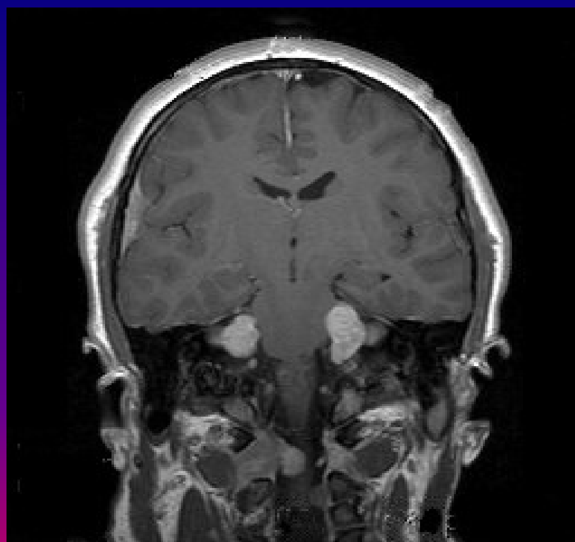
SCHWANNOMA

- Arises in cranial and spinal nerve roots
- 90% are in the 8th cranial nerve in the cerebello-pontine angle (= acoustic neuroma)
- Extra-axial, benign and slow-growing
- Well-circumscribed and encapsulated
- Symptoms are due to compression
- Can be usually completely removed
- “Fish flesh” gross appearance
- May be hemorrhagic



52 y o man. Diagnosis: Acoustic neuroma **=** Schwannoma. **Description:** Cellular Antoni A-areas with Verocay bodies (row of palisading = parallel nuclei) and less cellular Antoni B areas with myxoid change. **Stain:** HE.

NEUROFIBROMATOSIS 2 (NF2)



- **BILATERAL ACOUSTIC NEUROMAS (BANF) IN NF2**
- **THE PRODUCT OF *NF2* GENE <u>MERLIN</u> INHIBITS CELL PROLIFERATION AND CELL CYCLE PROGRESSION BY REPRESSING CYCLIN D1 PROMOTER ACTIVITY**

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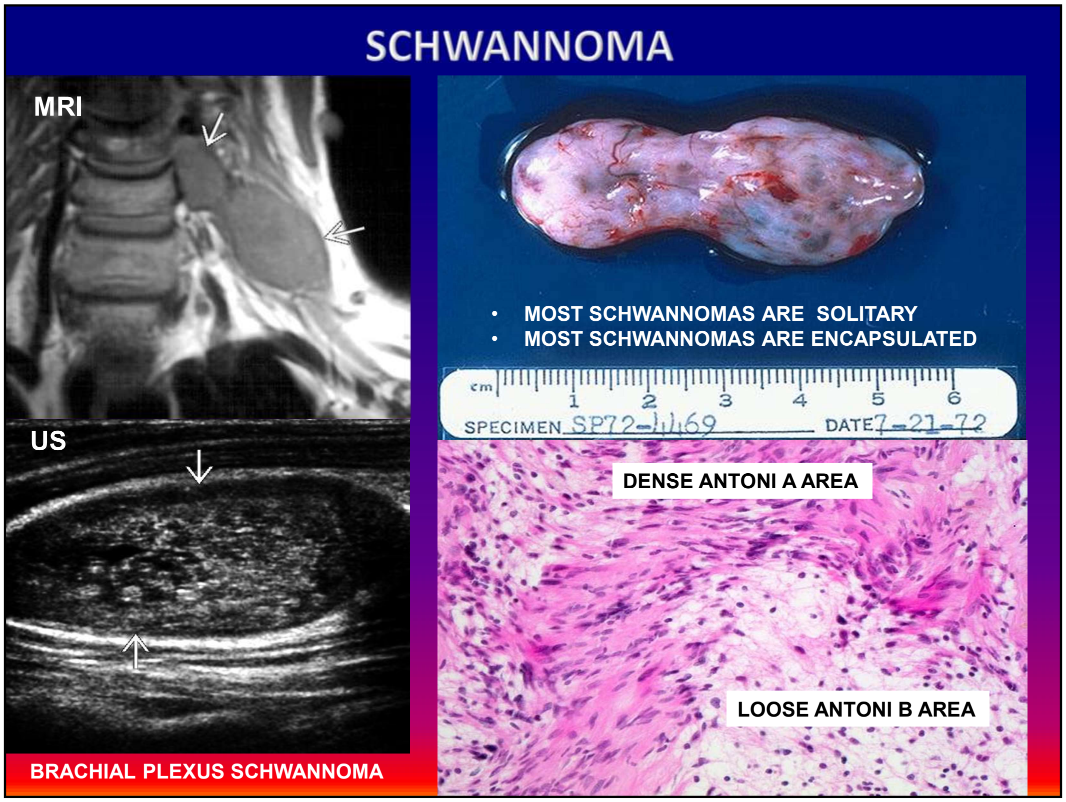
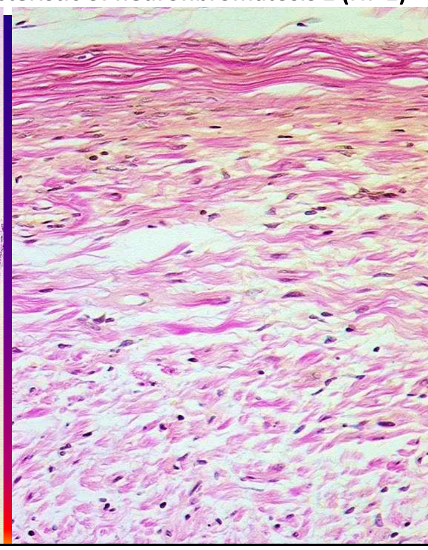


MRI and Ultrasound of a brachial plexus schwannoma

NEUROFIBROMA

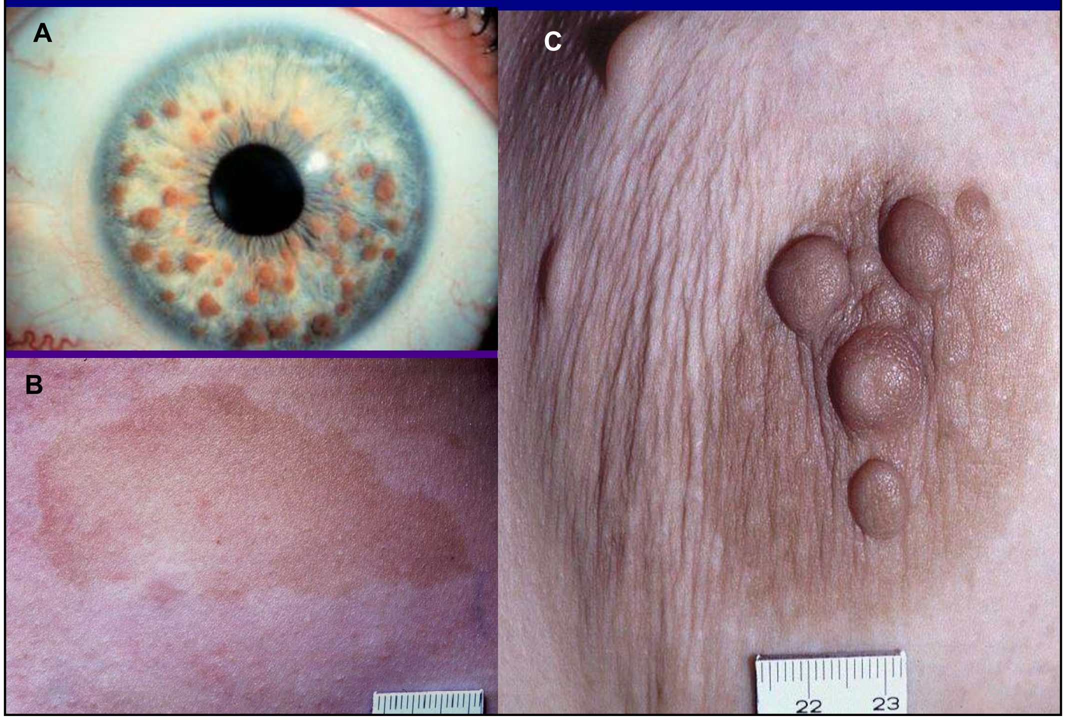
- Peripheral nerve tumor
- Proliferation of a mixture of Schwann cells and fibroblasts causes fusiform enlargement of the nerve
- Schwann cells and fibroblasts are arranged loosely in a wavy pattern
- Multiple neurofibromas involving long segments of peripheral nerves create plexiform structures characteristic of neurofibromatosis 1 (NF 1)



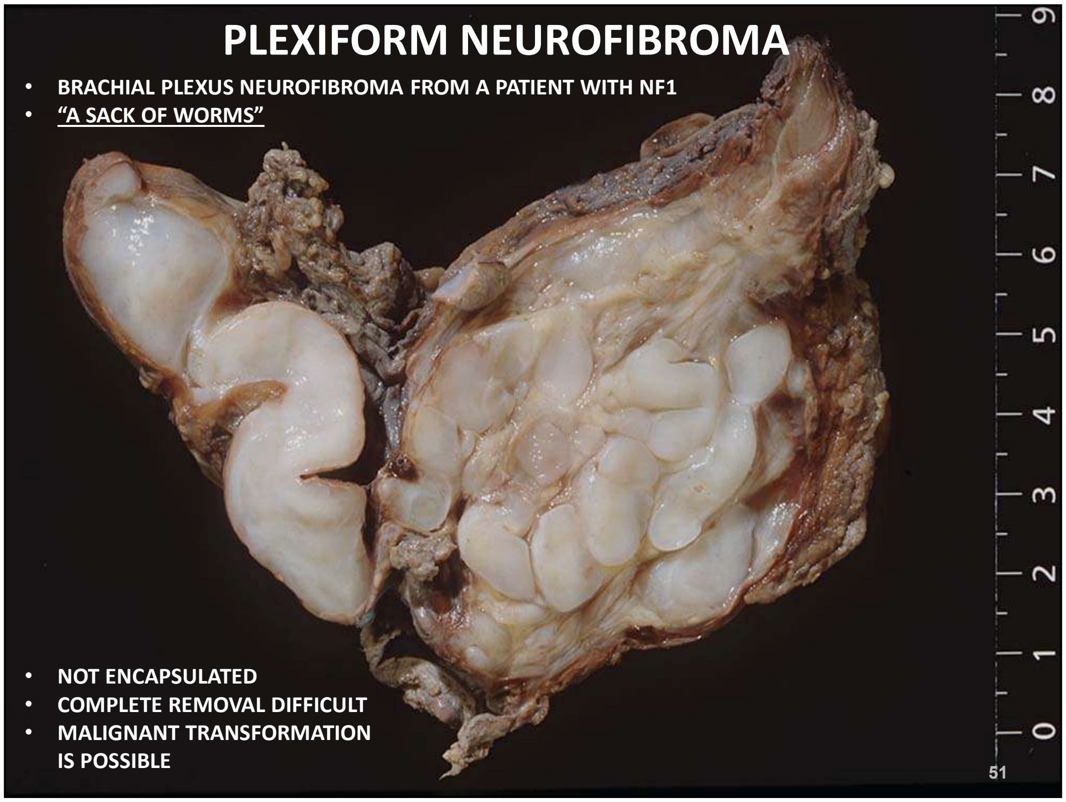
NEUROFIBROMATOSIS NF1

- Autosomal dominant disorder caused by mutations in the tumor suppressor neurofibromin, encoded on the long arm of chromosome 17 (17q).
- Neurofibromin is a negative regulator of the oncoprotein Ras
- Loss of neurofibromin function is a main feature of NF1-associated tumors.
- The single copy of normal *NF1* allele is mutated or silenced in tumors arising in the setting of NF1.
- Patients develop neurofibromas, malignant peripheral nerve sheath tumors, "optic gliomas," and other glial tumors.
- Patients with NF1 also have other problems:
  - learning disabilities,
  - Seizures
  - skeletal abnormalities
  - vascular abnormalities including arterial stenoses,
  - pigmented nodules of the iris (Lisch nodules*)*
  - pigmented skin lesions e.g. axillary freckling and café-au-lait spots.

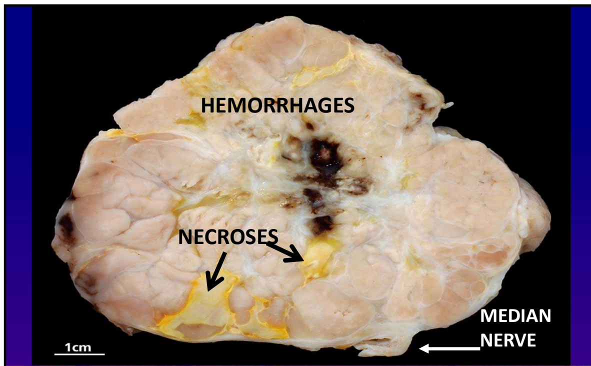
NEUROFIBROMATOSIS

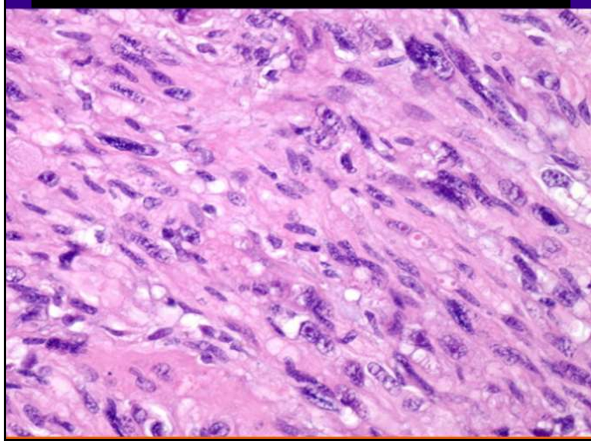


von Recklinghausen neurofibromatosis VRNF



17 y o female. Diagnosis: Plexiform neurofibroma of the brachial plexus. **Description:** Worm-like tumor masses. **Clinical findings:** Neurofibromatosis type 1. The patient has a mutation in the gene located in the long arm of chromosome 17 (17Q) encoding the tumor suppressor neurofibromin.





MALIGNANT PERIPHERAL NERVE SHEATH TUMOR

GENERAL:

- **Arises de novo or from transformation of a plexiform neurofibroma**
- **May develop as a complication of neurofibromatosis**
- **May be caused by radiation therapy**
- **Highly malignant behavior with local invasion and metastases**
- **Multiple recurrences common**

MORPHOLOGY:

- **Poorly defined tumor mass infiltrates along the parent nerve and into the surrounding tissues**
- **Highly cellular; resembles fibrosarcoma**
- **Composed of monomorphic highly atypical spindle cells arranged in interlacing fascicles, palisades and whorls**
- **High mitotic rate and nuclear atypia**
- **Tumor necrosis and hemorrhage are common**
- **Positive for s-100 (most)**

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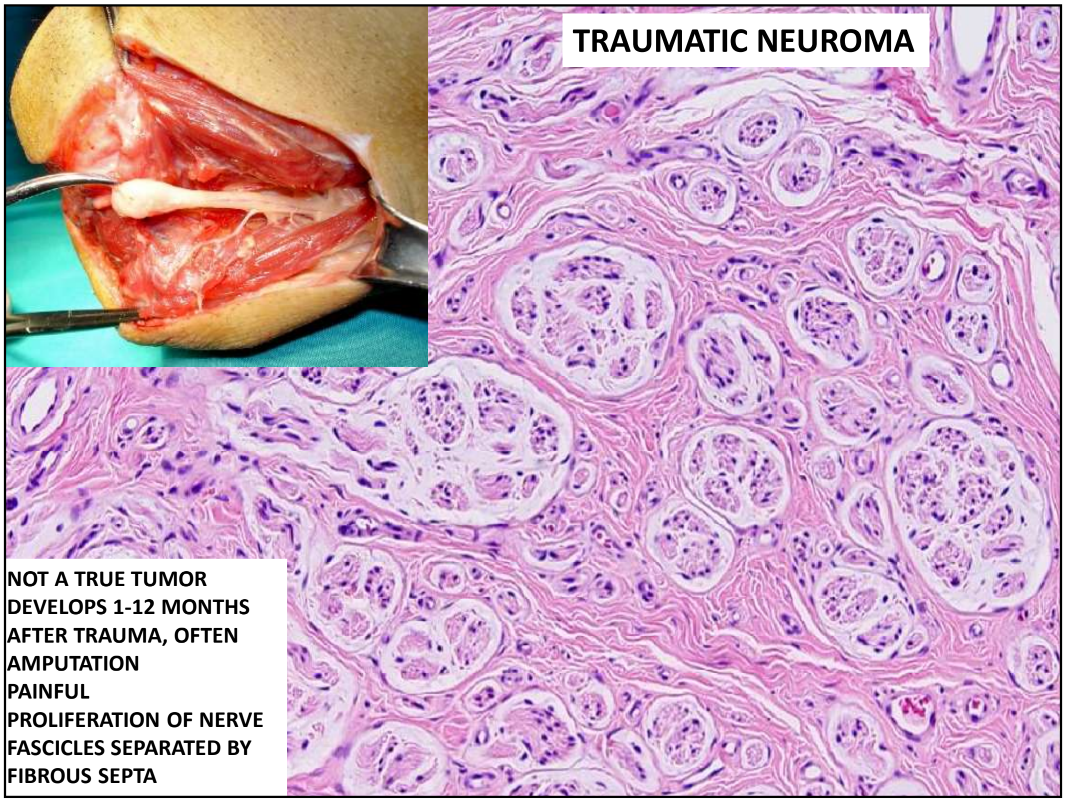
45 yof. Diagnosis: Malignant peripheral nerve sheath tumor. **Description:** Lobulated tumor with focal necrosis. **Clinical findings:** Tumor in the median nerve.

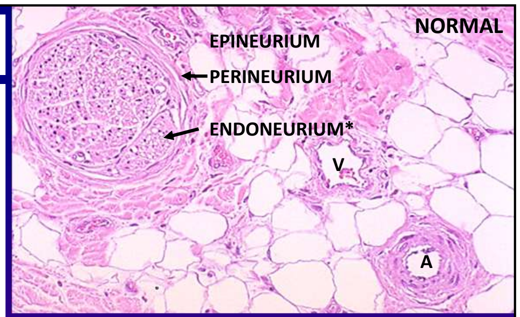
Genetics:

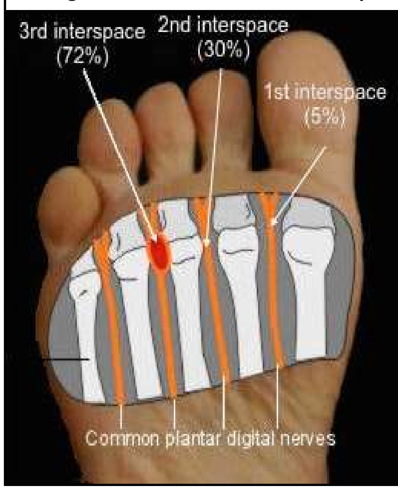


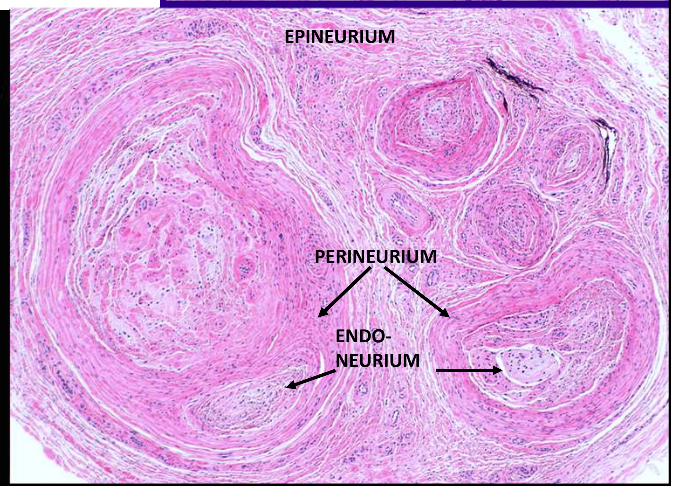
Courtesy of Jian-Hua Qiao

MORTON’S NEUROMA

- **AKA morton’s metatarsalgia, plantar neuroma, intermetatarsal neuroma, plantar interdigital neuroma**
- **Not a true tumor but a perineural fibrous proliferation**
- **Tight shoes (high heels) and gait abnormalities cause trauma and entrapment of common plantar digital nerves at the bifurcation point**







* Endoneurium is a layer of delicate connective tissue that encloses the myelin sheath of a nerve fiber and is made up of loose, supportive endoneurial connective tissue.

THE END

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